AutoDD: Learning to Select Drift Detectors for Evolving Data Streams

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***Abstract*—Concept drift, the change in the distribution of streaming data over time, significantly impacts the performance of machine learning models. While numerous drift detection methods have been proposed, selecting and configuring the most suitable detector for a specific data stream remains a challenging task. Existing AutoML frameworks for streaming data that automate model selection rely on fixed drift detection strategies, leaving a critical gap in automation. This paper introduces *AutoDD* (Automated Drift Detector), a novel framework that jointly automates detector selection and hyperparameter configuration using a hierarchical learning approach. AutoDD extracts comprehensive online meta-features and employs a two-level hierarchical selection algorithm to adaptively choose both the detector and its configuration. Experiments on ten synthetic and real-world datasets show that AutoDD outperforms state-of-the-art detectors while maintaining computational efficiency.**

***Index Terms*—concept drift detection, algorithm selection, contextual bandits, online meta-learning, data streams**

I. Introduction

Concept drift, where the statistical properties of streaming data evolve over time, poses a significant challenge in data stream mining [1], [2], incremental learning [3], and non-stationary learning [4]. As data distributions change, machine learning models experience degraded performance, necessitating timely detection and adaptation. This phenomenon manifests in various forms, including sudden (abrupt) drift, gradual drift, incremental drift, and recurring drift. For instance, an electricity price prediction model may encounter sudden drift when market regulations change abruptly, experience gradual drift as consumer behaviour shifts over seasons, or observe recurring drift driven by daily demand cycles [4].

Over the past two decades, numerous drift detection methods have been developed to address this challenge [1], [4], spanning error rate-based [5]–[7], distribution-based [8], [9], uncertainty-based [10], and meta-learning [11] paradigms. Yet no single detector performs best across all conditions [1], [12], because each embeds assumptions about how drift manifests. As a result, a detector well suited to abrupt changes may respond slowly to gradual ones [4], [13], and the most appropriate choice can shift as the stream itself evolves [12]. However, a fundamental challenge remains unresolved in systematically selecting and configuring the most appropriate drift detector for a given data stream.

Automated Machine Learning (AutoML) has revolutionised the field by addressing similar selection challenges in batch learning scenarios [14]. Recent advancements have extended AutoML to streaming data, enabling automatic selection and configuration for supervised learning tasks [15], [16]. However, these systems treat drift detection as a fixed component, requiring manual specification and leaving a critical gap in the automation framework.

Automating this choice online is challenging because drift labels are unavailable as the stream arrives [1], [4], so detector quality must be inferred from a proxy reward derived from downstream model behaviour rather than from ground-truth change points [6], [11], and because the joint space of detectors and their configurations is too large to search as a single flat action set [14]. Unlike detector ensembles that run and combine many detectors at once [17], [18], or meta-learning methods that learn a fixed mapping offline [11], an effective solution must select and configure one detector online and continually refine this choice as the stream evolves. To bridge this gap, we introduce *AutoDD* (Automated Drift Detector), a modular framework that automates drift detector selection and configuration using a hierarchical learning approach, compatible with any model or AutoML pipeline. Figure 1 gives an overview of the framework. AutoDD makes three main contributions.

- We propose *AutoDD*, a novel framework for automated drift detector selection and configuration in evolving data streams.
- AutoDD formalises the problem of joint detector and hyperparameter selection and solves it via a two-level hierarchical bandit using online meta-features, LinUCB [19] for detector selection, and Thompson Sampling [20] for hyperparameter configuration.
- A comprehensive experimental study across ten benchmarks shows that AutoDD ranks first on mean accuracy and macro-F1 against state-of-the-art detectors, while maintaining computational efficiency.

The remainder of this paper is organised as follows. Section II reviews related work, Section III defines the problem, and Section IV presents AutoDD and its theoretical analysis. Sections V and VI describe the experimental setup and results, while Sections VII and VIII discuss limitations and conclude the paper.

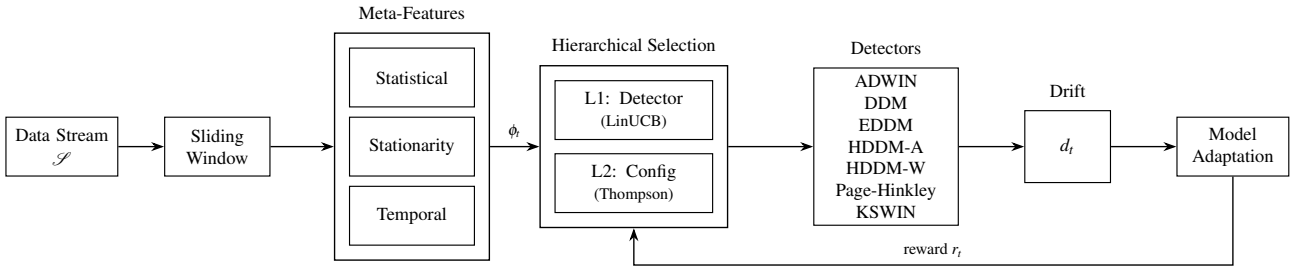


Fig. 1. Overview of the AutoDD framework. The incoming data stream $\mathcal{S}$ is buffered into a tumbling window, from which statistical, stationarity, and temporal meta-features form the context $\boldsymbol{\phi}_t$. Hierarchical selection then chooses a detector with LinUCB (Level 1) and a hyperparameter configuration with Thompson Sampling (Level 2) from the detector portfolio. The configured detector emits a drift signal d_t that triggers model adaptation, and the resulting reward r_t is fed back to update both selection levels.

II. Related Work

Drift detection methods can be broadly categorised into three paradigms. **Error rate-based methods** monitor fluctuations in classifier error rates, triggering alarms when changes are statistically significant. DDM [5] pioneered the warning-drift framework by monitoring prediction accuracy under binomial distribution assumptions, followed by EDDM [13] which focuses on distances between errors to detect gradual drift, HDDM [21] which applies Hoeffding's inequality with exponentially weighted moving averages, and ADWIN [6] which maintains adaptive sliding windows that automatically shrink when distribution changes are detected. Page-Hinkley [22] applies cumulative sum control charts to detect mean shifts in sequential data. Recent advances include MDDM [7], which uses McDiarmid's inequality for shorter detection delays with tighter concentration bounds, and ADDM [23], which models error rates as SETAR time series [24] to achieve lower false alarm rates. **Distribution-based methods** compare data distributions across windows by quantifying dissimilarity between new and historical data distributions [8], including approaches based on information theory [25], competence models [26], and the Kolmogorov-Smirnov test (KSWIN) [9] which detects changes in univariate feature distributions. KSWIN is a recent representative of the statistical two-sample test family, which also includes earlier detectors such as IKS [27] and WRS [17]. **Uncertainty and learning-based methods** address limitations of error-rate monitoring. PUDD [10] leverages prediction confidence for earlier detection by monitoring changes in prediction uncertainty distributions, while Meta-ADD [11] employs prototypical networks to classify drift types via meta-learning on diverse drift scenarios.

Recent developments in automated algorithm selection and hyperparameter optimisation (CASH) [14] have been extended to streaming scenarios [28]. However, these streaming AutoML systems automate supervised task selection while assuming a fixed, manually specified drift detector, so the detector itself is never selected or configured automatically. AutoDD addresses this gap by bringing the detector into the automation loop, learning online which detector and configuration best suit the current stream.

III. Problem Formulation

We consider a supervised data stream in which labelled instances $(\mathbf{x}_t, y_t)$ arrive sequentially, where $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ is the feature vector and y_t is the corresponding label. The underlying data distribution may evolve over time, resulting in concept drift when

$$P_t(\mathbf{x}, y) \neq P_{t'}(\mathbf{x}, y), \qquad t \neq t'. \tag{1}$$

Given a portfolio of drift detectors $\mathcal{D} = \{D_1, \ldots, D_K\}$, where each detector D_i has an associated hyperparameter space $\mathcal{H}_i$, AutoDD addresses the problem of dynamically selecting and configuring the most suitable detector for the current stream conditions. Inspired by the CASH formulation [14], the objective at time t is

$$(D_t^*, \boldsymbol{\eta}_t^*) = \arg \max_{D_i \in \mathcal{D},\, \boldsymbol{\eta} \in \mathcal{H}_i} \mathrm{Perf}(D_i(\boldsymbol{\eta}) \mid \boldsymbol{\phi}_t)\,, \tag{2}$$

where $\boldsymbol{\phi}_t$ represents the current stream context and $\mathrm{Perf}(\cdot)$ measures the resulting downstream predictive performance.

IV. Automated Drift Detector

A. Overview

AutoDD employs an adaptive closed-loop architecture to dynamically select and configure drift detectors. The system first initialises a portfolio of drift detectors and their candidate hyperparameter configurations. During online operation, meta-features describing the current stream characteristics are extracted from each incoming window. These features provide the context for a LinUCB agent, which selects a detector, while Thompson Sampling chooses a suitable hyperparameter configuration for that detector. The configured detector monitors the stream and triggers model adaptation when drift is detected. Performance feedback is then used to update both selection levels, allowing AutoDD to improve its decisions as the stream evolves. This hierarchical design separates detector selection from hyperparameter optimisation, avoiding a large flat search over all detector–configuration pairs. It also enables the framework to adapt both the detector family and its configuration as stream characteristics change over time. Algorithm 1 summarises this adaptive learning cycle.

Algorithm 1: AutoDD

Input: Stream $\mathcal{S}$, detector portfolio $\mathcal{D}$, window size w, exploration parameter α, regularisation λ, configurations per detector C, Beta prior (a_0, b_0)

Output: Drift signals d_t

Initialise $\mathbf{A}_i \leftarrow \lambda\mathbf{I}$ and $\mathbf{b}_i \leftarrow \mathbf{0}$ for each detector D_i
Initialise $(a_{i,c}, b_{i,c}) \leftarrow (a_0, b_0)$ for each configuration
Initialise predictive model $\mathcal{M}$ and window $W \leftarrow \emptyset$
foreach $(\mathbf{x}_t, y_t) \in \mathcal{S}$ **do**
  Add $(\mathbf{x}_t, y_t)$ to W
  if $|W| = w$ **then**
    $\boldsymbol{\phi}_t \leftarrow \textsc{Extract}(W)$
      `// extract stream meta-features`
    foreach *detector* $D_i \in \mathcal{D}$ **do**
      $\hat{\boldsymbol{\theta}}_i \leftarrow \mathbf{A}_i^{-1}\mathbf{b}_i$
      $\mathrm{UCB}_i \leftarrow \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha\sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1}\boldsymbol{\phi}_t}$
    $i_t \leftarrow \arg\max_i \mathrm{UCB}_i$
      `// LinUCB Operation`
    for $c = 1$ **to** C **do**
      $\psi_c \sim \mathrm{Beta}(a_{i_t,c}, b_{i_t,c})$
    $c_t \leftarrow \arg\max_c \psi_c$
      `// Thompson Sampling Operation`
    $d_t \leftarrow D_{i_t}(\boldsymbol{\eta}_{i_t,c_t}).\textsc{Detect}(W)$
      `// apply drift detector`
    if $d_t = 1$ **then**
      Reset predictive model $\mathcal{M}$
    Compute reward r_t using Eq. (8)
    $\mathbf{A}_{i_t} \leftarrow \mathbf{A}_{i_t} + \boldsymbol{\phi}_t\boldsymbol{\phi}_t^\top$
    $\mathbf{b}_{i_t} \leftarrow \mathbf{b}_{i_t} + r_t\boldsymbol{\phi}_t$
      `// update Level 1`
    $\tilde{r}_t \leftarrow (r_t + 1)/2$
    $a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t$
    $b_{i_t,c_t} \leftarrow b_{i_t,c_t} + 1 - \tilde{r}_t$
      `// update Level 2`
    $W \leftarrow \emptyset$

B. Framework Components

AutoDD performs one selection cycle for every tumbling window of w instances, with detector selection, configuration, drift detection, and feedback occurring at the end of each window.

Meta-Feature Extraction: The meta-feature set contains three categories. *Statistical* features capture velocity, dimensionality, sparsity, noise level, and class imbalance. *Stationarity* features capture error volatility, KL divergence, autocorrelation, mean shift, and variance shift. *Temporal* features capture trend, periodicity, change frequency, change magnitude, and recurrence. All features are computed online over the current window and normalised to $[0, 1]$, producing the context vector $\boldsymbol{\phi}_t$.

TABLE I
DETECTOR PORTFOLIO WITH HYPERPARAMETER CONFIGURATIONS.

Detector	Type	Parameter	Range	Scale	Default
ADWIN	Error (window)	δ_{adwin}	[0.0001, 0.1]	log	0.002
Page-Hinkley	Error (CUSUM)	θ_{ph}	[10, 100]	int	50
		δ_{ph}	[0.001, 0.01]	log	0.005
DDM	Error-based	w_{thresh}	[1.5, 3.0]	linear	2.0
		d_{thresh}	[2.0, 4.0]	linear	3.0
EDDM	Error-based	α_{eddm}	[0.90, 0.99]	linear	0.95
		β_{eddm}	[0.85, 0.95]	linear	0.90
HDDM-A	Error-based	δ_{hddm}	[0.0001, 0.01]	log	0.001
HDDM-W	Error-based	λ_{hddm}	[0.01, 0.1]	log	0.05
KSWIN	Distribution-based	α_{ks}	[0.001, 0.05]	log	0.005
		w_{ks}	[50, 500]	int	100

Detector Portfolio: AutoDD maintains a diverse portfolio of drift detectors spanning error-based and distribution-based approaches. Table I summarises their hyperparameters, search ranges, sampling scales, and literature defaults. Each detector maintains C candidate configurations. The first uses the literature default, while the remaining configurations are sampled from the specified ranges using logarithmic, linear, or integer sampling as appropriate.

C. Hierarchical Learning Approach

AutoDD decomposes detector adaptation into two levels rather than treating every detector–configuration pair as an independent action. Level 1 selects the detector according to the current stream context, while Level 2 tunes the selected detector's hyperparameters.

Level 1 – Detector Selection via LinUCB: Detector selection is formulated as a contextual multi-armed bandit problem, where the stream meta-features $\boldsymbol{\phi}_t$ form the context and each detector corresponds to an arm. LinUCB [19] selects the detector with the largest upper confidence bound:

$$i_t = \arg\max_i \left(\boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha\sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1}\boldsymbol{\phi}_t} \right), \tag{3}$$

where $\hat{\boldsymbol{\theta}}_i = \mathbf{A}_i^{-1}\mathbf{b}_i$ represents the estimated relationship between the stream context and the reward obtained by detector i. The first term favours detectors with high estimated reward, while the second encourages exploration of detectors with greater uncertainty. The parameter α controls the exploration–exploitation trade-off.

After observing reward r_t, the selected detector is updated as

$$\mathbf{A}_{i_t} \leftarrow \mathbf{A}_{i_t} + \boldsymbol{\phi}_t\boldsymbol{\phi}_t^\top, \qquad \mathbf{b}_{i_t} \leftarrow \mathbf{b}_{i_t} + r_t\boldsymbol{\phi}_t. \tag{4}$$

Level 2 – Hyperparameter Selection via Thompson Sampling: After selecting detector D_{i_t}, Thompson Sampling [20] chooses among its candidate hyperparameter configurations. Each configuration maintains a Beta distribution with parameters $(a_{i,c}, b_{i,c})$. At each selection step,

$$\psi_c \sim \mathrm{Beta}(a_{i_t,c}, b_{i_t,c}), \tag{5}$$

and the configuration with the largest sampled value is selected:

$$c_t = \arg\max_c \psi_c. \tag{6}$$

After receiving reward r_t, it is normalised to $\tilde{r}_t = (r_t+1)/2$ and the selected configuration is updated as

$$a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t, \qquad b_{i_t,c_t} \leftarrow b_{i_t,c_t} + 1 - \tilde{r}_t. \tag{7}$$

This hierarchical design allows detector selection and configuration to adapt jointly while avoiding a flat search over every possible detector–configuration pair.

D. Evaluation and Adaptation

Because ground-truth drift locations are generally unavailable in real-world streams, AutoDD uses changes in downstream predictive accuracy as a proxy for detector quality. Let Δa_t denote the accuracy drop between consecutive windows. The reward is defined as

$$r_t = \begin{cases} \rho_{\mathrm{tp}} + \min(\kappa \Delta a_t, \rho_{\mathrm{tp}}) & d_t = 1,\ \Delta a_t > \tau_{\mathrm{sig}}, \\ -\rho_{\mathrm{fp}} & d_t = 1,\ \Delta a_t \le \tau_{\mathrm{sig}}, \\ -\rho_{\mathrm{fn}} - \min(\Delta a_t, \rho_{\mathrm{fn}}) & d_t = 0,\ \Delta a_t > \tau_{\mathrm{sig}}, \\ \rho_{\mathrm{tn}} & d_t = 0,\ \Delta a_t \le \tau_{\mathrm{sig}}. \end{cases} \tag{8}$$

Here, d_t indicates whether the selected detector raises an alarm and τ_{sig} determines whether the observed accuracy drop is considered significant. The reward terms ρ_{tp}, ρ_{fp}, ρ_{fn}, and ρ_{tn} correspond to the four possible alarm outcomes, while κ controls the contribution of the observed accuracy drop. The reward is bounded to support stable bandit updates. When drift is detected, the downstream model is reset and resumes incremental learning. In the deployed implementation, Level 1 is additionally updated whenever drift is detected, while Level 2 is updated at window boundaries.

E. Computational Complexity

Most meta-features require linear-time statistics over the current window, while periodicity uses an FFT-based computation. Consequently, the per-window time and space complexities are

$$\begin{aligned} T_{\mathrm{win}} &= O(w d_{\mathrm{in}} + w \log w + K m^2), \\ S_{\mathrm{win}} &= O(K m^2 + w d_{\mathrm{in}}). \end{aligned} \tag{9}$$

where w is the window size, d_{in} is the input dimensionality, K is the number of detectors, and m is the number of meta-features. The runtime is primarily determined by meta-feature extraction and LinUCB updates, while memory is dominated by the LinUCB matrices and the window buffer.

F. Theory

We analyse the two selection levels of AutoDD separately under standard contextual and stochastic bandit assumptions. These results characterise the idealised proxy-selection problems and do not constitute a regret guarantee for the complete deployed system.

Assumptions. For the Level 1 analysis, we make the following standard assumptions [29], [30]:

- **(A1) Bounded context:** The normalised meta-feature vector satisfies $\|\boldsymbol{\phi}_t\|_2 \le \sqrt{m}$.
- **(A2) Linear expected reward:** For each detector i, there exists a bounded parameter vector $\boldsymbol{\theta}_i$ such that

$$\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i. \tag{10}$$

- **(A3) Bounded reward noise:** The deviation of the observed reward from its conditional expectation is conditionally sub-Gaussian. This is consistent with the bounded proxy reward used by AutoDD.

The accuracy-based reward in Eq. (8) additionally assumes that larger proxy rewards correspond, on average, to better detector behaviour. This proxy-fidelity condition motivates the optimisation objective but is not required for the regret bounds below.

Remark on concept drift. Although the stream distribution changes over time, A2 does not require the stream itself to be stationary. It assumes only that changes relevant to detector performance are sufficiently represented by the context $\boldsymbol{\phi}_t$, so that the mapping from context to expected reward remains approximately stable.

Theorem 1 (Level 1 Proxy Regret). Assume A1–A3 and fix the configuration of each detector. Using the LinUCB selection rule in Eq. (3), with a standard confidence-calibrated exploration parameter α and regularisation $\lambda > 0$, the cumulative Level 1 regret over T selection rounds satisfies, with high probability,

$$\begin{aligned} R_1(T) &= \sum_{t=1}^{T} \left(\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t} \right) \\ &= \tilde{O}\left(m\sqrt{KT} \right). \end{aligned} \tag{11}$$

Proof Sketch. Standard self-normalised concentration results for linear contextual bandits [29] give a confidence region around each estimate $\hat{\boldsymbol{\theta}}_i$. On this event, the instantaneous regret of the detector selected by LinUCB is bounded by

$$2\alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}}. \tag{12}$$

Summing these terms over time and applying the elliptical potential lemma [31] to the K detector estimators gives

$$R_1(T) = \tilde{O}\left(m\sqrt{KT} \right). \qquad \square$$

Proposition 2 (Level 2 Proxy Regret). Consider the idealised Level 2 problem in which the C configurations of each detector produce independent bounded rewards from stationary distributions. Under the standard Bernoulli reduction, Thompson Sampling [32] gives the aggregate expected regret

$$\mathbb{E}[R_2(T)] = O\left(\sqrt{CKT \log(1+T)} \right). \tag{13}$$

Proof Sketch. Conditioned on a detector being selected, its configuration search is a C-armed stochastic bandit. The reward is mapped to $[0,1]$ using $\tilde{r}_t = (r_t + 1)/2$, after which the standard Thompson Sampling regret bound applies. Summing the regret across the K detector-specific configuration problems and using the fact that their total number of selections is T gives the stated bound.

The implemented update in Eq. (7) uses the fractional reward $\tilde{r}_t$ directly rather than drawing a Bernoulli observation. Therefore, this result should be interpreted as a surrogate bound for the corresponding idealised Thompson Sampling problem rather than a formal guarantee for the deployed update. □

The two results analyse detector selection and hyperparameter selection separately and therefore do not establish regret against a joint detector–configuration oracle. If the two surrogate objectives are treated as separable, their indicative combined order is

$$R_{\text{proxy}}(T) \lesssim \tilde{O}\left(m\sqrt{KT} + \sqrt{CKT}\right). \tag{14}$$

This expression is reported only as an indicative proxy rate for the hierarchical selection process, not as a guarantee for the complete AutoDD system.

V. Experiments

A. Datasets and Base Models

We evaluate AutoDD on ten benchmark datasets comprising five synthetic and five real-world streams, with characteristics detailed in Table II. To isolate drift detection performance from model selection effects, all methods use two fixed base models, GaussianNB and Hoeffding Adaptive Tree (HAT) [33]. These two learners represent fundamentally different model families (probabilistic and tree-based) and are the standard benchmark base learners used in the data stream mining literature [2], [17].

B. Baselines

We compare AutoDD against twelve baselines, including a no-detector baseline and seven portfolio detectors: ADWIN, DDM, EDDM, HDDM-A, HDDM-W, Page-Hinkley, and KSWIN. We further include four state-of-the-art methods: MDDM [7], ADDM [23], PUDD [10], and Meta-ADD [11]. All detectors are evaluated using the hyperparameter settings recommended in the literature. To maintain a diverse yet manageable action space, the portfolio includes one representative from each detector family rather than every available variant. Since AutoDD is detector-agnostic, expanding the portfolio changes only the set of candidate actions and does not alter the underlying method.

TABLE II
Dataset characteristics (Multiple* = Sudden, Gradual, Incremental, and Recurring drift types)

Dataset	Type	Instances	Features	Drift Type
SEA	Synthetic	500,000	3	Multiple*
HYP	Synthetic	500,000	10	Multiple*
AGR	Synthetic	100,000	9	Multiple*
RBF	Synthetic	200,000	10	Multiple*
RTG	Synthetic	100,000	30	Multiple*
Elec	Real-world	45,312	8	Unknown
Weather	Real-world	18,159	8	Unknown
Spam	Real-world	4,601	57	Unknown
Airlines	Real-world	539,383	7	Unknown
Poker	Real-world	829,201	10	Unknown

C. Evaluation Protocol

We evaluate predictive performance using prequential accuracy under the standard test-then-train protocol [2]. For the synthetic streams, we additionally report prequential macro-averaged F1 to account for class imbalance. Because some synthetic streams exhibit continuous drift without discrete ground-truth drift locations, detector-level matching metrics cannot be defined consistently and are therefore not reported. For the real-world streams, we instead report the number of detected drifts (DN) as an indicator of adaptation frequency.

Accuracy reflects overall predictive performance but can be dominated by the majority class, whereas macro-F1 gives equal importance to all classes and is therefore more sensitive to minority-class degradation caused by delayed or excessive adaptation. Reporting both metrics provides a more complete view of predictive performance under drift. Neither metric directly measures drift-detection timing, which remains a limitation of proxy-based evaluation.

Following established benchmarking guidance [12], DN is interpreted relative to the range observed across methods, since unusually low or high values may indicate insufficient or excessive adaptation. This protocol enables a consistent comparison across methods without requiring unavailable ground-truth drift annotations.

D. Implementation Details

AutoDD uses all m meta-features with a fixed window size, bandit parameters, and C configurations per detector throughout. Results are averaged over 10 runs with random seeds uniformly sampled from the range [1, 100]. Implementation uses Python with River [34] on a MacBook Pro with Apple M2 Pro and 16GB RAM. All code and datasets are publicly available.[1]

VI. Results

We conduct a comprehensive evaluation of AutoDD against classical detectors and state-of-the-art methods.

A. Synthetic Datasets

Tables III and IV compare all methods on the five synthetic benchmarks. AutoDD achieves the best macro-F1 on every synthetic stream under both base models and the best mean accuracy under GaussianNB, with Meta-ADD ranking second across most datasets. The advantage is not uniform on accuracy. Under both base models, Meta-ADD is more accurate on RBF and RTG, the two highest-dimensional streams, though AutoDD retains the better macro-F1 on both. Under HAT these two streams give Meta-ADD the higher mean accuracy. The advantage is largest on datasets with abrupt drift, where adaptive detector switching is most beneficial, and narrows on gradual drift where Meta-ADD's learned representations are competitive. Figures 2a–2b further illustrate the prequential behaviour on SEA, showing how performance evolves as the stream changes.

[1] https://github.com/TechyNilesh/AutoDD

TABLE III

SYNTHETIC DATASETS UNDER GAUSSIANNB. RANK IN (·). **BOLD** = BEST, UNDERLINE = SECOND-BEST PER COLUMN.

Method	SEA		HYP		AGR		RBF		RTG		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
No Detector	83.8 (11)	0.40 (10)	76.4 (11)	0.34 (8)	**78.8 (1)**	0.41 (10)	50.4 (13)	0.09 (10)	58.2 (11)	0.10 (7)	69.5 (12)	0.27 (10)
ADWIN	84.0 (10)	0.61 (9)	81.9 (9)	0.00 (9)	71.5 (10)	0.51 (7)	69.6 (10)	0.00 (11)	64.5 (9)	0.00 (8)	74.3 (10)	0.22 (13)
DDM	83.8 (11)	0.00 (12)	86.1 (7)	0.63 (7)	71.6 (9)	0.49 (9)	71.3 (4)	0.64 (4)	64.4 (10)	0.00 (8)	75.4 (7)	0.35 (9)
EDDM	84.3 (9)	0.40 (10)	85.7 (8)	0.63 (7)	65.3 (11)	0.24 (11)	67.7 (12)	0.38 (9)	65.0 (8)	0.40 (5)	73.6 (11)	0.41 (8)
HDDM-A	84.6 (7)	0.69 (6)	81.5 (10)	0.00 (9)	72.2 (8)	0.63 (6)	69.7 (9)	0.00 (11)	64.4 (10)	0.00 (8)	74.5 (9)	0.26 (11)
HDDM-W	84.4 (8)	0.66 (7)	86.6 (6)	0.71 (4)	73.1 (6)	0.67 (5)	70.1 (6)	0.51 (8)	65.8 (6)	0.56 (4)	76.0 (6)	0.62 (6)
Page-Hinkley	84.7 (6)	0.70 (5)	81.5 (10)	0.00 (9)	71.5 (10)	0.49 (9)	69.4 (11)	0.00 (11)	64.5 (9)	0.00 (8)	74.3 (10)	0.24 (12)
KSWIN	83.7 (12)	0.35 (11)	85.7 (8)	0.65 (6)	71.5 (10)	0.50 (8)	70.0 (7)	0.61 (6)	65.2 (7)	0.39 (6)	75.2 (8)	0.50 (7)
MDDM	86.7 (4)	0.65 (8)	87.2 (5)	0.68 (5)	72.3 (7)	0.71 (4)	69.8 (8)	0.56 (7)	66.1 (5)	0.56 (4)	76.4 (5)	0.63 (5)
ADDM	87.7 (3)	0.83 (3)	88.7 (4)	0.73 (3)	73.6 (5)	0.67 (5)	70.9 (5)	0.62 (5)	67.6 (4)	0.71 (3)	77.7 (4)	0.71 (4)
PUDD	86.2 (5)	0.81 (4)	89.5 (3)	0.84 (2)	73.7 (4)	0.72 (3)	73.4 (3)	0.72 (2)	67.8 (3)	0.71 (3)	78.1 (3)	0.76 (3)
Meta-ADD	88.9 (2)	0.87 (2)	90.8 (2)	**0.91 (1)**	74.1 (3)	0.79 (2)	**78.0 (1)**	0.70 (3)	**72.9 (1)**	0.76 (2)	80.9 (2)	0.81 (2)
AutoDD	**93.1 (1)**	**0.90 (1)**	**94.6 (1)**	**0.91 (1)**	78.5 (2)	**0.84 (1)**	74.3 (2)	**0.81 (1)**	69.2 (2)	**0.84 (1)**	**81.9 (1)**	**0.86 (1)**

TABLE IV

SYNTHETIC DATASETS UNDER HAT. RANK IN (·). **BOLD** = BEST, UNDERLINE = SECOND-BEST PER COLUMN.

Method	SEA		HYP		AGR		RBF		RTG		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
No Detector	86.2 (10)	0.42 (9)	79.1 (13)	0.36 (10)	81.3 (2)	0.43 (11)	53.1 (12)	0.11 (8)	60.8 (13)	0.12 (10)	72.1 (13)	0.29 (11)
ADWIN	88.1 (8)	0.64 (7)	85.5 (12)	0.06 (11)	74.6 (10)	0.55 (8)	73.3 (10)	0.07 (9)	68.3 (11)	0.11 (11)	78.0 (11)	0.29 (11)
DDM	88.0 (9)	0.10 (10)	89.0 (7)	0.67 (8)	74.7 (9)	0.52 (10)	74.6 (5)	0.69 (3)	68.1 (12)	0.13 (9)	78.9 (7)	0.42 (9)
EDDM	88.0 (9)	0.46 (8)	88.6 (8)	0.68 (7)	69.2 (12)	0.29 (12)	71.8 (11)	0.42 (7)	68.8 (9)	0.47 (7)	77.3 (12)	0.46 (8)
HDDM-A	88.3 (6)	0.64 (7)	86.8 (10)	0.06 (11)	75.1 (7)	0.63 (7)	73.8 (7)	0.07 (9)	69.1 (7)	0.11 (11)	78.6 (9)	0.30 (10)
HDDM-W	88.4 (5)	0.65 (6)	89.5 (6)	0.72 (6)	75.9 (5)	0.67 (6)	73.9 (6)	0.53 (6)	69.7 (6)	0.58 (6)	79.5 (6)	0.63 (6)
Page-Hinkley	88.2 (7)	0.65 (6)	86.0 (11)	0.06 (11)	74.8 (8)	0.52 (10)	73.5 (9)	0.07 (9)	68.5 (10)	0.11 (11)	78.2 (10)	0.28 (12)
KSWIN	88.0 (9)	0.42 (9)	88.3 (9)	0.66 (9)	74.5 (11)	0.53 (9)	73.7 (8)	0.63 (4)	68.9 (8)	0.43 (8)	78.7 (8)	0.53 (7)
MDDM	89.1 (4)	0.67 (5)	90.3 (5)	0.75 (5)	75.5 (6)	0.75 (4)	73.8 (7)	0.60 (5)	69.8 (5)	0.61 (5)	79.7 (5)	0.68 (5)
ADDM	89.8 (3)	0.86 (3)	91.9 (4)	0.77 (4)	76.6 (4)	0.72 (5)	75.0 (4)	0.69 (3)	71.3 (3)	0.77 (3)	80.9 (4)	0.76 (4)
PUDD	89.8 (3)	0.83 (4)	92.6 (2)	0.83 (3)	77.1 (3)	0.76 (3)	76.4 (2)	0.74 (2)	71.9 (2)	0.74 (4)	81.6 (3)	0.78 (3)
Meta-ADD	90.9 (2)	0.91 (2)	92.5 (3)	0.92 (2)	77.1 (3)	0.81 (2)	**84.8 (1)**	0.74 (2)	**80.4 (1)**	0.79 (2)	**85.1 (1)**	0.83 (2)
AutoDD	**96.7 (1)**	**0.93 (1)**	**98.4 (1)**	**0.93 (1)**	**82.4 (1)**	**0.90 (1)**	75.8 (3)	**0.84 (1)**	70.9 (4)	**0.88 (1)**	84.8 (2)	**0.90 (1)**

TABLE V

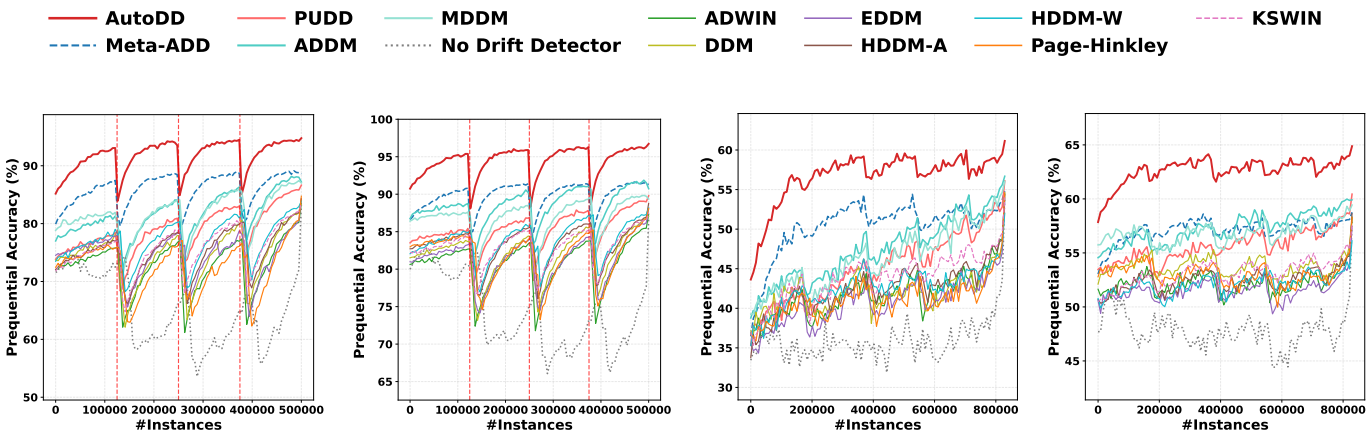
REAL-WORLD DATASETS UNDER GNB. RANK IN (·). **BOLD** = BEST, UNDERLINE = SECOND-BEST PER COLUMN. DN IS REPORTED WITHOUT RANKING.

Method	Elec		Weather		Spam		Airlines		Poker		Mean	
	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN
No Detector	73.2 (13)	0	68.9 (12)	0	77.2 (12)	0	59.3 (13)	0	54.6 (6)	0	66.6 (13)	0
ADWIN	79.2 (11)	22	69.3 (11)	7	92.1 (9)	2	62.5 (7)	76	54.3 (7)	3	71.5 (10)	22
DDM	82.3 (8)	22	70.6 (9)	5	91.8 (11)	2	61.8 (9)	20	54.2 (8)	3	72.1 (8)	10.4
EDDM	82.4 (7)	22	73.8 (6)	9	92.9 (5)	2	61.3 (11)	182	53.6 (10)	4	72.8 (7)	43.8
HDDM-A	82.7 (6)	22	72.2 (8)	9	92.5 (8)	2	62.6 (6)	114	54.6 (6)	6	72.9 (6)	30.6
HDDM-W	81.9 (9)	22	72.6 (7)	9	91.9 (10)	1	61.0 (12)	269	52.3 (11)	394	71.9 (9)	139
Page-Hinkley	78.0 (12)	16	69.4 (10)	6	91.8 (11)	1	62.8 (5)	64	54.3 (7)	17	71.3 (12)	20.8
KSWIN	81.1 (10)	22	68.0 (13)	9	92.6 (7)	2	61.6 (10)	269	53.7 (9)	266	71.4 (11)	113.6
MDDM	83.9 (5)	22	74.8 (5)	9	92.8 (6)	2	62.2 (8)	110	56.1 (4)	190	74.0 (5)	66.6
ADDM	84.8 (3)	22	76.3 (2)	9	93.9 (3)	2	63.9 (3)	74	56.5 (2)	177	75.1 (2)	56.8
PUDD	84.4 (4)	22	75.4 (3)	9	93.2 (4)	2	64.0 (2)	117	56.2 (3)	81	74.6 (4)	46.2
Meta-ADD	85.4 (2)	22	74.9 (4)	9	94.5 (2)	2	63.8 (4)	139	55.5 (5)	182	74.8 (3)	70.8
AutoDD	**89.6 (1)**	22	**80.0 (1)**	9	**97.9 (1)**	2	**68.2 (1)**	118	**59.7 (1)**	188	**79.1 (1)**	67.8

TABLE VI

REAL-WORLD DATASETS UNDER HAT. RANK IN (·). **BOLD** = BEST, UNDERLINE = SECOND-BEST PER COLUMN. DN IS REPORTED WITHOUT RANKING.

Method	Elec		Weather		Spam		Airlines		Poker		Mean	
	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN
No Detector	76.5 (13)	0	71.4 (12)	0	80.1 (12)	0	62.1 (12)	0	56.9 (11)	0	69.4 (12)	0
ADWIN	82.4 (11)	22	73.1 (10)	9	94.6 (8)	2	66.3 (5)	76	57.7 (8)	3	74.8 (9)	22.4
DDM	85.5 (7)	22	73.5 (9)	7	93.9 (11)	2	65.2 (9)	20	58.0 (6)	3	75.2 (8)	10.8
EDDM	85.2 (8)	22	77.1 (6)	9	94.8 (6)	2	64.6 (11)	182	57.8 (7)	4	75.9 (6)	43.8
HDDM-A	86.3 (6)	22	75.8 (8)	9	94.7 (7)	2	66.1 (6)	114	58.5 (5)	6	76.3 (5)	30.6
HDDM-W	85.1 (9)	22	76.0 (7)	9	94.5 (9)	1	64.9 (10)	269	56.8 (12)	404	75.5 (7)	141
Page-Hinkley	81.7 (12)	16	72.8 (11)	6	94.2 (10)	1	66.5 (4)	64	57.5 (9)	17	74.5 (11)	20.8
KSWIN	84.3 (10)	22	71.2 (13)	9	94.9 (5)	2	65.4 (8)	269	57.0 (10)	266	74.6 (10)	113.6
MDDM	86.5 (5)	22	78.0 (5)	9	94.6 (8)	2	66.0 (7)	110	60.0 (2)	190	77.0 (4)	66.6
ADDM	87.3 (4)	22	79.2 (2)	9	95.8 (3)	2	66.9 (3)	74	59.6 (4)	177	77.8 (2)	56.8
PUDD	87.5 (3)	22	78.6 (3)	9	95.6 (4)	2	67.7 (2)	117	59.7 (3)	81	77.8 (2)	46.2
Meta-ADD	87.9 (2)	22	78.2 (4)	9	95.9 (2)	2	66.5 (4)	139	59.6 (4)	182	77.6 (3)	70.8
AutoDD	**93.8 (1)**	22	**85.1 (1)**	9	**99.0 (1)**	2	**72.6 (1)**	118	**65.1 (1)**	188	**83.1 (1)**	67.8



(a) SEA (GNB) (b) SEA (HAT) (c) Poker (GNB) (d) Poker (HAT)

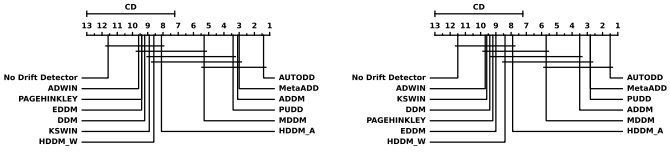
Fig. 2. Prequential accuracy over time. SEA (synthetic) and Poker (real-world) datasets with GaussianNB and HAT base models.

B. Real-world Datasets

Tables V and VI report results on real-world benchmarks where drift patterns are unknown. AutoDD ranks first on mean accuracy under both base models, with the improvement consistent across all five datasets. Detection counts remain within the consensus range of the compared methods, avoiding unusually low or high detection frequencies. On Elec, Weather, and Spam, AutoDD reports the same detection count as the strongest comparators, so its accuracy gain comes from when it adapts rather than from detecting more often. Figures 2c and 2d confirm that AutoDD tracks or leads the baselines on the Poker stream and recovers promptly after accuracy drops.

C. Statistical Significance

Figures 3a and 3b show Critical Difference diagrams from the Friedman–Nemenyi test ($\alpha = 0.05$). AutoDD achieves the best mean rank under both base models, and the No Detector baseline ranks last, confirming that drift detection is beneficial across these benchmarks. AutoDD is separated from every fixed portfolio detector by more than the critical difference, while Meta-ADD, PUDD, ADDM, and MDDM fall within the same clique. The consistent ranking across GaussianNB and HAT indicates that AutoDD's advantage is not tied to a particular downstream learner.



(a) GaussianNB (b) HAT

Fig. 3. Critical Difference diagrams (Friedman–Nemenyi, $\alpha = 0.05$). AutoDD achieves the best average rank under both base models.

D. Ablation Study

Table VII reports performance when each component is removed in turn, using GaussianNB averaged across the Elec, Weather, Airlines, and Poker streams. Removing the Level 1 LinUCB detector selector causes the largest accuracy drop, confirming that detector-stream matching is the primary driver of adaptive performance. Removing Level 2 Thompson Sampling produces a smaller but meaningful degradation, and all three meta-feature categories provide complementary gains, with statistical features contributing most. Figure 4 illustrates the resulting behaviour on the HYP stream. The LinUCB layer continuously re-selects detectors as meta-features evolve, keeping the windowed error bounded and enabling rapid recovery after detected drifts.

TABLE VII
ABLATION STUDY RESULTS (ACCURACY AND MACRO-F1 AVERAGED ACROSS REAL-WORLD DATASETS UNDER GAUSSIANNB).

Configuration	Accuracy	Macro-F1	ΔAcc
Full AutoDD	**74.5%**	**0.60**	–
w/o Temporal	67.9%	0.54	-6.5%
w/o Statistical	64.9%	0.52	-9.6%
w/o Stationarity	67.5%	0.54	-7.0%
w/o LinUCB (Level 1)	56.6%	0.45	-17.9%
w/o Thompson (Level 2)	69.2%	0.55	-5.3%

E. Sensitivity Study

Figure 5 shows AutoDD's sensitivity to its three key hyperparameters. Performance peaks at $w = 100$, where smaller windows introduce high variance and larger windows delay drift response. The exploration parameter α, fixed at 1.0 in all experiments, shows a clear optimum, with under-exploration causing exploitation bias and over-exploration reducing accuracy. The regularisation parameter λ is robust across the tested range, and AutoDD degrades gracefully around the default settings. The same pattern is observed across synthetic and real-world streams, indicating that the selected defaults are not driven by a single dataset. Overall, window size has the strongest influence on performance, while the bandit hyperparameters are less sensitive.

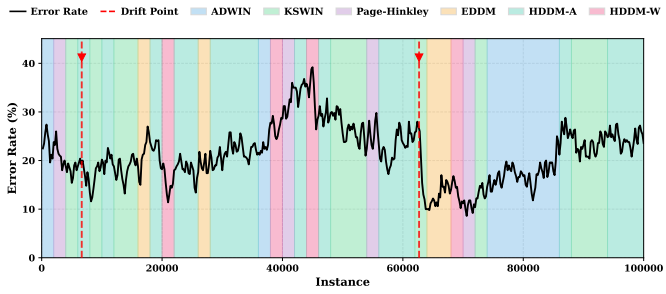
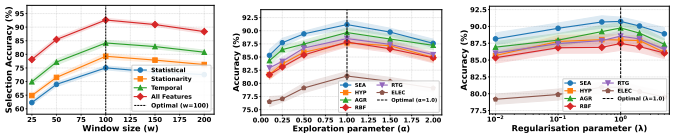


Fig. 4. Detector selection by AutoDD over the first 100,000 instances of the HYP stream. All quantitative results use the full 500,000-instance stream. Coloured bands show the dominant LinUCB-selected detector per block, the black curve the windowed prequential error rate, and dashed red lines the drift alarms raised by AutoDD, not ground-truth drift points.

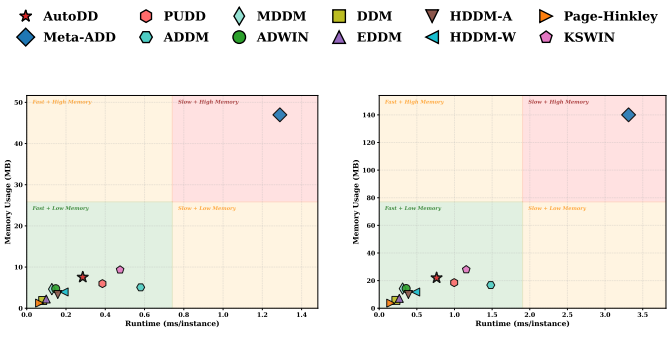


(a) w (b) α (c) λ

Fig. 5. Hyperparameter sensitivity analysis showing prequential accuracy across different values of (a) window size w, (b) exploration parameter α, and (c) regularisation parameter λ.

F. Computational Efficiency

Figures 6a and 6b place all methods in the runtime-memory plane. AutoDD occupies the fast, low-memory region under both base models, with the hierarchical selection mechanism introducing only modest overhead. Most classical detectors also remain in this region, whereas Meta-ADD incurs substantially higher runtime and memory costs. AutoDD remains close to the lightweight fixed detectors despite performing online detector and configuration selection. It is strictly more efficient than Meta-ADD and PUDD while attaining the better mean rank, making it practical for high-velocity streams.



(a) GaussianNB (b) HAT

Fig. 6. Runtime vs. memory trade-off. AutoDD lies in the Fast + Low Memory quadrant under both base models.

VII. Discussion and Limitations

AutoDD addresses automated detector and hyperparameter selection, but several limitations remain. First, the window size is fixed throughout the stream, although the sensitivity analysis identifies it as the most influential setting. While the chosen default performs consistently across our benchmarks, adapting the window size online could further improve responsiveness to changing stream dynamics. Second, the reward is based on downstream accuracy and therefore couples detector quality with the behaviour of the base learner. Window-level smoothing and the significance threshold reduce this dependence, and the consistency observed across the two base learners provides empirical support for the proxy. Nevertheless, a fully label-free reward would provide a cleaner measure of detector quality. Third, our experiments use GaussianNB and HAT with full model resets after detected drift, leaving richer models and softer adaptation strategies, such as warm-starting, selective resets, replay, or ensemble updates, for future work.

The comparison also has two asymmetries that should be considered when interpreting the performance margins. The strongest comparators, MDDM, ADDM, PUDD, and Meta-ADD, are evaluated as baselines but are not included in AutoDD's selectable portfolio, which is restricted to the detectors in Table I. Moreover, the baselines use fixed literature-recommended hyperparameters, whereas AutoDD adapts detector configurations online. Consequently, its gains may reflect both detector selection and hyperparameter adaptation. Future evaluation with broader portfolios and equivalent tuning budgets would help separate these effects. Finally, AutoDD is designed for tabular data streams, while image, audio, and text streams remain outside the scope of the present study [35].

VIII. Conclusion and Future Work

This work advances streaming AutoML by automating the selection and configuration of drift detectors, a component that is typically specified manually. We formalised the joint choice of detector and hyperparameters as a contextual selection problem and introduced AutoDD, which solves it online with a two-level hierarchical bandit. Level 1 uses LinUCB over online meta-features to select a detector, and Level 2 uses Thompson Sampling to select its configuration, both driven by a bounded proxy reward because no drift labels are available at run time. Across ten benchmarks and two base learners, AutoDD attains the best mean accuracy and macro-F1 and the best average rank under the Friedman–Nemenyi test, while the ablation confirms detector selection as the main driver and its runtime and memory stay in the fast, low-memory region.

Several directions follow from these limitations. The most immediate is learning the window size w in an outer adaptive loop, since the sensitivity analysis identifies it as the most influential setting. A second is replacing the accuracy-based proxy with a label-free reward that decouples detector credit from the downstream learner. Finally, evaluating on streams with annotated change points would allow detection delay and false alarms to be scored directly.

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APPENDIX

TABLE VIII
ACCURACY (%) BROKEN DOWN BY DRIFT TYPE UNDER GAUSSIANNB. AUTODD LEADS ON SUDDEN, INCREMENTAL, AND RECURRING DRIFT, WHILE META-ADD EDGES AHEAD ON GRADUAL DRIFT OWING TO ITS LEARNED DRIFT-TYPE REPRESENTATIONS.

Method	Sudden	Gradual	Incremental	Recurring
ADWIN	82.4	71.3	68.5	75.2
DDM	79.8	74.6	72.1	71.8
EDDM	75.2	78.4	74.8	69.3
HDDM-A	81.7	72.1	69.8	74.5
HDDM-W	80.1	76.2	73.5	74.6
Page-Hinkley	81.9	70.8	68.2	75.8
KSWIN	76.5	77.8	75.2	72.1
MDDM	81.3	75.8	74.1	73.9
ADDM	82.9	77.2	75.6	74.8
PUDD	82.1	79.5	76.8	75.4
Meta-ADD	84.6	**82.3**	78.9	77.8
AutoDD	**88.2**	81.7	**80.4**	**79.6**

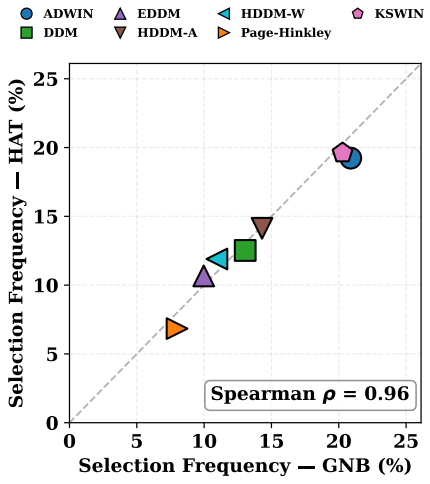


Fig. 7. Mean detector selection frequency under GaussianNB against HAT across all ten datasets. Every detector lies near the diagonal (Spearman $\rho = 0.96$, $p < 0.001$), indicating that the proxy reward tracks detector quality rather than base-learner artefacts.

TABLE IX
DETECTOR SELECTION FREQUENCY (%) BY STREAM CHARACTERISTIC. THE LEARNED POLICY ALIGNS WITH DOMAIN KNOWLEDGE, FAVOURING ADWIN AND DDM ON HIGH-VELOCITY STREAMS AND KSWIN ON HIGH-NOISE AND HIGH-DIMENSIONAL STREAMS.

Detector	High Velocity	High Noise	High Dim.
ADWIN	28.4	12.3	15.7
DDM	22.1	8.5	11.2
EDDM	8.3	18.7	14.5
HDDM-A	15.2	10.4	12.8
HDDM-W	12.6	14.2	13.1
Page-Hinkley	5.8	8.1	7.2
KSWIN	7.6	27.8	25.5

TABLE X
MEAN DETECTION DELAY (INSTANCES) ON THE SYNTHETIC DATASETS WITH DISCRETE CHANGE POINTS, LOWER IS BETTER. AUTODD ATTAINS THE LOWEST DELAY OF ALL COMPARATORS, SELECTING RESPONSIVE DETECTORS DURING DRIFT-PRONE WINDOWS WHILE AVOIDING OVER-SENSITIVE CHOICES IN STABLE PERIODS.

Method	SEA	HYP	AGR	RBF	Mean
ADWIN	45.2	78.3	52.1	81.4	64.3
DDM	38.7	62.5	48.9	65.2	53.8
EDDM	52.3	71.8	61.4	74.6	65.0
HDDM-A	43.8	75.1	50.3	78.6	62.0
HDDM-W	41.5	58.2	45.7	62.8	52.1
Page-Hinkley	46.9	80.5	53.8	83.2	66.1
KSWIN	49.1	68.7	58.2	71.4	61.9
MDDM	39.4	55.8	44.2	60.5	50.0
ADDM	36.5	51.3	41.6	57.8	46.8
PUDD	32.8	45.2	37.5	51.6	41.8
Meta-ADD	35.2	48.6	39.8	55.3	44.7
AutoDD	**28.4**	**42.1**	**34.2**	**48.7**	**38.4**

TABLE XI
AUTODD PROXY-ORACLE SELECTION AGREEMENT (%) BY STREAM QUARTER. AGREEMENT RISES MONOTONICALLY FROM THE FIRST TO THE FINAL QUARTER ON EACH DATASET SHOWN, SHOWING THAT THE LINUCB LAYER LEARNS RELIABLE DETECTOR SELECTIONS FROM THE REWARD SIGNAL AS THE STREAM PROGRESSES.

Dataset	0–25%	25–50%	50–75%	75–100%
SEA	71.2	85.4	91.8	94.3
HYP	68.5	82.1	89.6	93.1
AGR	65.8	79.3	87.2	91.5
Elec	72.4	84.7	90.3	92.8
Poker	69.1	81.6	88.9	93.6
Mean	69.4	82.6	89.6	93.1

TABLE XII
AUTODD SETTINGS USED IN ALL EXPERIMENTS. DETECTOR-SPECIFIC HYPERPARAMETER RANGES AND DEFAULTS ARE GIVEN IN TABLE I.

Component	Parameter	Value
Stream	Window size w	100
	Meta-feature dimensionality m	15
Portfolio	Number of detectors K	7
	Configurations per detector C	5
Level 1 (LinUCB)	Exploration α	1.0
	Regularisation λ	1.0
Level 2 (Thompson)	Beta prior (a_0, b_0)	$(1, 1)$
Reward (Eq. (8))	Significance threshold τ_{sig}	0.05
	True-positive base ρ_{tp}	0.5
	Accuracy-drop scale κ	2
	False-positive penalty ρ_{fp}	0.3
	False-negative base ρ_{fn}	0.5
	True-negative reward ρ_{tn}	0.1
Evaluation	Runs per dataset	10
	Random seeds	$\mathcal{U}\{1, \ldots, 100\}$

TABLE XIII
THE 15 ONLINE META-FEATURES FORMING THE CONTEXT VECTOR ϕ_t, COMPUTED OVER EACH TUMBLING WINDOW OF SIZE w.

Category	Feature	Formula/Description
Statistical	Velocity	$w/\Delta t_w$
	Dimensionality	d_{in}
	Sparsity	$\frac{1}{w d_{\text{in}}} \sum_{i=1}^{w} \sum_{j=1}^{d_{\text{in}}} \mathbb{I}[\mathbf{x}_{i,j} = 0]$
	Noise level	$\frac{1}{d_{\text{in}}} \sum_{j=1}^{d_{\text{in}}} \sigma_j / \text{range}_j$
	Class imbalance	$\max_y n_y / \min_y n_y$
Stationarity	Error volatility	$\sqrt{\frac{1}{w} \sum_{i=1}^{w} (e_i - \bar{e})^2}$
	KL divergence	$D_{\text{KL}}(\hat{P}_t \Vert \hat{P}_{t-1})$ (empirical window distributions)
	Autocorrelation	$\text{corr}(e_{2:w}, e_{1:w-1})$
	Mean shift	$\frac{1}{d_{\text{in}}} \sum_{j=1}^{d_{\text{in}}} \lvert\mu_{t,j} - \mu_{t-1,j}\rvert$
	Variance shift	$\frac{1}{d_{\text{in}}} \sum_{j=1}^{d_{\text{in}}} \lvert\sigma^2_{t,j} - \sigma^2_{t-1,j}\rvert$
Temporal	Trend	$\text{Cov}(i, e_i)/\text{Var}(i)$
	Periodicity	$\arg\max_f \lvert\mathcal{F}\{e_i\}(f)\rvert$
	Change frequency	$\frac{1}{w} \sum_{i=2}^{w} \mathbb{I}[\lvert\Delta e_i\rvert > \tau_{\Delta e}]$, $\tau_{\Delta e} = \sigma_e/2$
	Change magnitude	$\frac{1}{\lvert\mathcal{C}_t\rvert} \sum_{i \in \mathcal{C}_t} \lvert\Delta e_i\rvert$, $\mathcal{C}_t = \{i : \lvert\Delta e_i\rvert > \tau_{\Delta e}\}$
	Recurrence score	$\text{corr}(e_{1:w/2}, e_{w/2+1:w})$